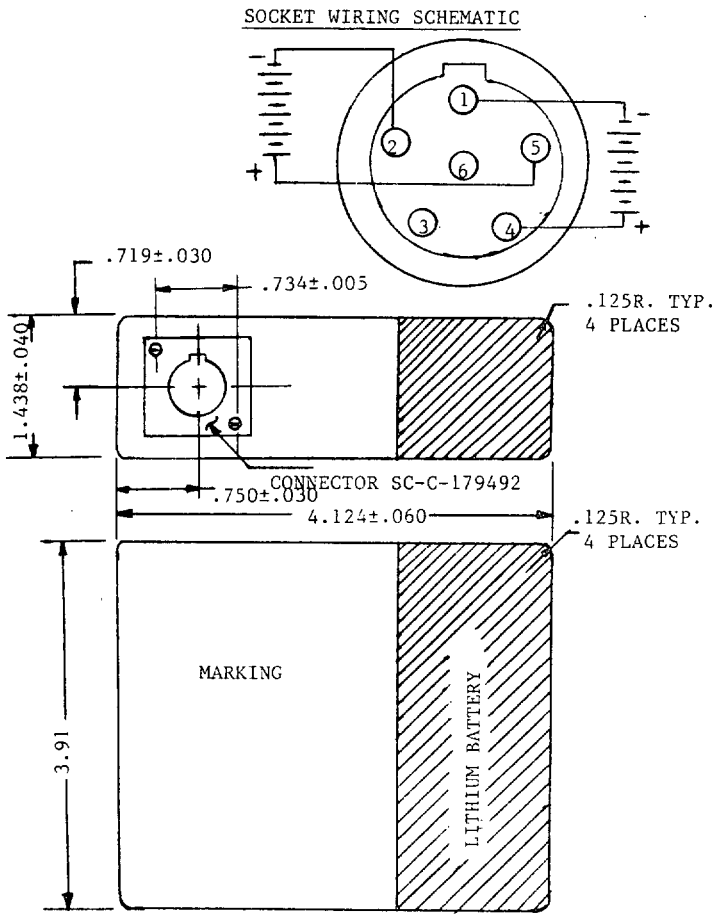
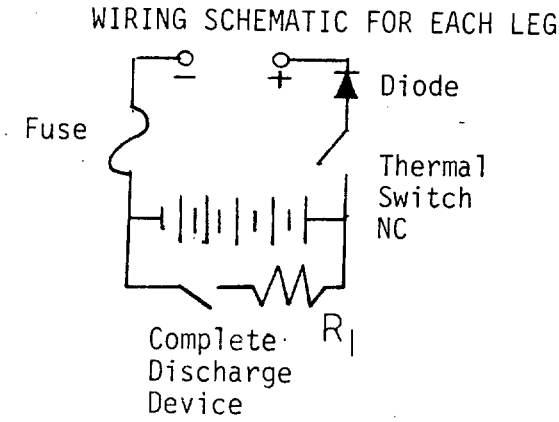
MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5557/U

This specification sheet is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the lithium sulfur dioxide non-rechargeable battery described herein shall consist of this document and the latest issue of MIL-B-49430(ER).





INCHES	MM
0.005	0.13
0.030	0.76
0.040	1.02
0.060	1.52
0.125	3.18
0.719	18.26
0.734	18.64
0.750	19.05
1.438	36.53
3.910	99.31
4.124	104.75

Figure 1. Battery dimensions and wiring schematic.

AMSC N/A FSC 6135

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NOTES

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are ±.030 inch.

3. The outer jacket and all battery components exterior to cells shall be fabricated from non-flammable and waterproof materials.

4. Socket wiring schematic shall be printed on jacket in suitable location near the socket on the terminal face of the battery.

5. The socket (connector) shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 0.0625 inchs below adjacent outside of the jacket when used initially, during, and after subsequent insertions of the mating plug.

6. R_1 value shall be between 460 ohms and 650 ohms and have a power rating of 0.5 watt.

REQUIREMENTS:

Dimension and configuration: See Figure 1.

Maximum voltage: 30.50 volts.

Minimum voltage (for capacity): 20.00 volts
(Consisting of two (2) each identical 12 volt sections.)

Coulombic ratio: Each cell shall have a coulombic ratio of lithium to sulfur dioxide of 0.95±0.1.

The battery shall contain an 8/10 ampere non-replaceable, slow blow fuse for each 12V leg wired in series with the cells in the negative leg.

The battery shall contain a diode wired in series in each 12 volt section. See paragraph 3.5.5.2 of the Basic Specification.

Terminals: Six (6) hole socket type with no obstruction of any holes (see Figure 1).

Weight (maximum): 1.1 pound (500 grams)

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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 20.00 Volts
I	3.0
L	1.5
H	5.5
HT	5.3
LT	1.3
T	2.8
D	2.8
ILR	1000

Initial voltage delay: During the I, H, HT, T, D, and ILR discharge tests, initial voltages below the 20.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 20.00 volts cannot exceed a 60.0 second duration.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery in each leg as described in paragraphs 3.5.5 and 3.5.5.1 of the Basic Specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para. 3.5.5.3 of the basic specification.

METHOD OF EXAMINATION AND TESTS:

Capacity tests:

1. Storage: Details on storage conditions for all specified capacity tests are described in the basic specification.(see 4.8.9).

2. Discharge:

Cells: Discharge at 0.65 amps constant current at 70±5°F for a minimum of 2.7 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 0.32 amperes constant current for 5.4 hours into voltage reversal.

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Batteries: For all capacity test except the H, HT, and ILR, the batteries shall be discharged at 0.65 amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through 0.65 amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. For the I capacity test the batteries shall be force discharged at 0.32 amperes constant current for 6.0 hours immediately after zero volts is obtained.

There shall be no venting, fire, or explosion when the batteries are force discharged. For the ILR test a fixed resistance of 13,350 ohms shall be used to a cut-off voltage of 20.00 volts or 1,000 hours.

Closed circuit voltage: Load resistance of 25.0 ohms shall be used between socket holes 2 and 5 and between socket holes 1 and 4. Voltage shall be above 10.00 volts within five seconds.

Custodian:
Army-ER

Preparing activity:
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